$$\mathbf{x}_d = \begin{bmatrix} x_T \\ y_T \\ z_T \\ \theta_a \end{bmatrix}$$

$$L_1 = 89.45 \text{ mm}, \qquad L_2 = 100 \text{ mm}, \qquad L_m = 35 \text{ mm}$$

$$L_3 = 100 \text{ mm}, \qquad L_4 = 129.15 \text{ mm}$$

$$L_r = \sqrt{L_2^2 + L_m^2}$$

$$L_r = \sqrt{(100)^2 + (35)^2}$$

$$\boxed{L_r = 105.9481 \text{ mm}}$$

$$\beta = \operatorname{atan2}(L_m, L_2)$$

$$\beta = \operatorname{atan2}(35, 100)$$

$$\boxed{\beta = 0.336675 \text{ rad} = 19.2900^\circ}$$

$$\psi = \frac{\pi}{2} - \beta$$

$$\psi = \frac{\pi}{2} - 0.336675$$

$$\boxed{\psi = 1.234122 \text{ rad} = 70.7100^\circ}$$

$$\theta_{2,\mathrm{DH}} = q_2 - \psi$$

$$\theta_{3,\mathrm{DH}} = q_3 + \psi$$

$$\theta_a = q_2 + q_3 + q_4 + \frac{\pi}{2}$$

$$\boxed{q_4 = \theta_a - q_2 - q_3 - \frac{\pi}{2}}$$

$$q_1 = \operatorname{atan2}(y_T, x_T)$$

$$\boxed{q_1 = \operatorname{atan2}(y_T, x_T)}$$

$$\mathbf{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} = \begin{bmatrix} \cos q_1 \sin\theta_a \\ \sin q_1 \sin\theta_a \\ \cos\theta_a \end{bmatrix}$$

$$\sqrt{a_x^2 + a_y^2 + a_z^2} = 1$$

$$\mathbf{p}_T = \begin{bmatrix} x_T \\ y_T \\ z_T \end{bmatrix}$$

$$\mathbf{w} = \mathbf{p}_T - L_4\mathbf{a}$$

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} x_T \\ y_T \\ z_T \end{bmatrix} - L_4 \begin{bmatrix} \cos q_1 \sin\theta_a \\ \sin q_1 \sin\theta_a \\ \cos\theta_a \end{bmatrix}$$

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} x_T - 129.15 \cos q_1 \sin\theta_a \\ y_T - 129.15 \sin q_1 \sin\theta_a \\ z_T - 129.15 \cos\theta_a \end{bmatrix}$$

$$\boxed{x_w = x_T - 129.15 \cos q_1 \sin\theta_a}$$

$$\boxed{y_w = y_T - 129.15 \sin q_1 \sin\theta_a}$$

$$\boxed{z_w = z_T - 129.15 \cos\theta_a}$$

$$r = \sqrt{x_w^2 + y_w^2}$$

$$r = \sqrt{(x_T - 129.15 \cos q_1 \sin\theta_a)^2 + (y_T - 129.15 \sin q_1 \sin\theta_a)^2}$$

$$h = z_w - L_1$$

$$h = z_T - 129.15 \cos\theta_a - 89.45$$

$$\boxed{h = z_T - 89.45 - 129.15 \cos\theta_a}$$

$$c = \sqrt{r^2 + h^2}$$

$$c = \sqrt{x_w^2 + y_w^2 + (z_w - 89.45)^2}$$

$$|L_r - L_3| \leq c \leq L_r + L_3$$

$$|105.9481 - 100| \leq c \leq 105.9481 + 100$$

$$\boxed{5.9481 \leq c \leq 205.9481 \text{ mm}}$$

$$u_\phi = \frac{c^2 - L_3^2 - L_r^2}{-2L_r L_3}$$

$$u_\phi = \frac{c^2 - (100)^2 - (105.9481)^2}{-2(105.9481)(100)}$$

$$-1 \leq u_\phi \leq 1$$

$$\phi = \arccos(u_\phi)$$

$$\boxed{\phi = \arccos\left[\frac{c^2 - (100)^2 - (105.9481)^2}{-2(105.9481)(100)}\right]}$$

$$\gamma = \operatorname{atan2}(h, r)$$

$$\boxed{\gamma = \operatorname{atan2}\left(z_T - 89.45 - 129.15\cos\theta_a, \sqrt{x_w^2 + y_w^2}\right)}$$

$$u_\alpha = \frac{L_3^2 - L_r^2 - c^2}{-2L_r c}$$

$$u_\alpha = \frac{(100)^2 - (105.9481)^2 - c^2}{-2(105.9481)c}$$

$$-1 \leq u_\alpha \leq 1$$

$$\alpha = \arccos(u_\alpha)$$

$$\boxed{\alpha = \arccos\left[\frac{(100)^2 - (105.9481)^2 - c^2}{-2(105.9481)c}\right]}$$

$$q_2^{(u)} = \frac{\pi}{2} - \beta - \alpha - \gamma$$

$$q_2^{(u)} = \frac{\pi}{2} - 0.336675 - \alpha - \gamma$$

$$\boxed{q_2^{(u)} = 1.234122 - \alpha - \gamma}$$

$$q_3^{(u)} = \pi - \psi - \phi$$

$$q_3^{(u)} = \pi - 1.234122 - \phi$$

$$\boxed{q_3^{(u)} = 1.907471 - \phi}$$

$$q_4^{(u)} = \theta_a - q_2^{(u)} - q_3^{(u)} - \frac{\pi}{2}$$

$$\boxed{q_4^{(u)} = \theta_a - q_2^{(u)} - q_3^{(u)} - \frac{\pi}{2}}$$

$$\boxed{\mathbf{q}^{(u)} = \begin{bmatrix} q_1 \\ q_2^{(u)} \\ q_3^{(u)} \\ q_4^{(u)} \end{bmatrix}}$$

$$q_2^{(d)} = \frac{\pi}{2} - (\gamma - \alpha + \beta)$$

$$q_2^{(d)} = \frac{\pi}{2} - \gamma + \alpha - 0.336675$$

$$\boxed{q_2^{(d)} = 1.234122 - \gamma + \alpha}$$

$$q_3^{(d)} = -\pi + (\phi - \psi)$$

$$q_3^{(d)} = -\pi + \phi - 1.234122$$

$$\boxed{q_3^{(d)} = \phi - 4.375714}$$

$$q_4^{(d)} = \theta_a - q_2^{(d)} - q_3^{(d)} - \frac{\pi}{2}$$

$$\boxed{q_4^{(d)} = \theta_a - q_2^{(d)} - q_3^{(d)} - \frac{\pi}{2}}$$

$$\boxed{\mathbf{q}^{(d)} = \begin{bmatrix} q_1 \\ q_2^{(d)} \\ q_3^{(d)} \\ q_4^{(d)} \end{bmatrix}}$$

$$\mathrm{wrap}(q_i) = \mathrm{atan2}\left(\sin q_i, \cos q_i\right)$$

$$\widetilde{\mathbf{q}}^{(u)} = \begin{bmatrix} \mathrm{wrap}(q_1^{(u)}) \\ \mathrm{wrap}(q_2^{(u)}) \\ \mathrm{wrap}(q_3^{(u)}) \\ \mathrm{wrap}(q_4^{(u)}) \end{bmatrix}$$

$$\widetilde{\mathbf{q}}^{(d)} = \begin{bmatrix} \mathrm{wrap}(q_1^{(d)}) \\ \mathrm{wrap}(q_2^{(d)}) \\ \mathrm{wrap}(q_3^{(d)}) \\ \mathrm{wrap}(q_4^{(d)}) \end{bmatrix}$$

$$\mathbf{q}_{\min} = \begin{bmatrix} -180^\circ \\ -111^\circ \\ -121^\circ \\ -100^\circ \end{bmatrix}$$

$$\mathbf{q}_{\max} = \begin{bmatrix} 180^\circ \\ 107^\circ \\ 92^\circ \\ 123^\circ \end{bmatrix}$$

$$\mathcal{S}_{\mathrm{v}} = \left\{ \widetilde{\mathbf{q}}^{(s)} \ : \ \mathbf{q}_{\min} \le \widetilde{\mathbf{q}}^{(s)} \le \mathbf{q}_{\max}, \quad s \in \{u, d\} \right\}$$

$$\mathbf{q}_c = \begin{bmatrix} q_{1c} \\ q_{2c} \\ q_{3c} \\ q_{4c} \end{bmatrix}$$

$$\Delta q_i^{(s)} = \mathrm{atan2}\left[\sin\left(\widetilde{q}_i^{(s)} - q_{ic} \right), \cos\left(\widetilde{q}_i^{(s)} - q_{ic} \right) \right]$$

$$\Delta\mathbf{q}^{(s)} = \begin{bmatrix} \Delta q_1^{(s)} \\ \Delta q_2^{(s)} \\ \Delta q_3^{(s)} \\ \Delta q_4^{(s)} \end{bmatrix}$$

$$J_s = \sqrt{ \left(\Delta\mathbf{q}^{(s)} \right)^T \Delta\mathbf{q}^{(s)} }$$

$$\boxed{ \mathbf{q}_{\text{seleccionada}} = \arg\min_{\widetilde{\mathbf{q}}^{(s)} \in \mathcal{S}_{\mathrm{v}}} J_s }$$

$$c < |L_r - L_3| \quad \vee \quad c > L_r + L_3 \quad \Longrightarrow \quad \mathcal{S}_{\mathrm{v}} = \varnothing$$

$$\mathcal{S}_{\mathrm{v}} = \varnothing \quad \Longrightarrow \quad \nexists \, \mathbf{q}_{\text{seleccionada}}$$

$$\boxed{ q_1 = \mathrm{atan2}(y_T, x_T) }$$

$$\boxed{\begin{aligned} q_1^{(u)} &= \operatorname{atan2}(y_T, x_T) \\ q_2^{(u)} &= \frac{\pi}{2} - \beta - \alpha - \gamma \\ q_3^{(u)} &= \pi - \psi - \phi \\ q_4^{(u)} &= \theta_a - q_2^{(u)} - q_3^{(u)} - \frac{\pi}{2} \end{aligned}}$$

$$\boxed{\begin{aligned} q_1^{(d)} &= \operatorname{atan2}(y_T, x_T) \\ q_2^{(d)} &= \frac{\pi}{2} - (\gamma - \alpha + \beta) \\ q_3^{(d)} &= -\pi + (\phi - \psi) \\ q_4^{(d)} &= \theta_a - q_2^{(d)} - q_3^{(d)} - \frac{\pi}{2} \end{aligned}}$$

$$\boxed{\begin{aligned} q_1^{(u)} &= \operatorname{atan2}(y_T, x_T) \\ q_2^{(u)} &= 1.234122 - \alpha - \gamma \\ q_3^{(u)} &= 1.907471 - \phi \\ q_4^{(u)} &= \theta_a - q_2^{(u)} - q_3^{(u)} - \frac{\pi}{2} \end{aligned}}$$

$$\boxed{\begin{aligned} q_1^{(d)} &= \operatorname{atan2}(y_T, x_T) \\ q_2^{(d)} &= 1.234122 - \gamma + \alpha \\ q_3^{(d)} &= \phi - 4.375714 \\ q_4^{(d)} &= \theta_a - q_2^{(d)} - q_3^{(d)} - \frac{\pi}{2} \end{aligned}}$$

$$\boxed{\mathbf{q}^{(u)} = \begin{bmatrix} \operatorname{atan2}(y_T, x_T) \\ \frac{\pi}{2} - \beta - \alpha - \gamma \\ \pi - \psi - \phi \\ \theta_a - q_2^{(u)} - q_3^{(u)} - \frac{\pi}{2} \end{bmatrix}}$$

$$\boxed{\mathbf{q}^{(d)} = \begin{bmatrix} \operatorname{atan2}(y_T, x_T) \\ \frac{\pi}{2} - (\gamma - \alpha + \beta) \\ -\pi + (\phi - \psi) \\ \theta_a - q_2^{(d)} - q_3^{(d)} - \frac{\pi}{2} \end{bmatrix}}$$